

Effective Programming Practices for Economists

Scientific Computing

Broadcasting

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Examples of broadcasting

```
>>> a = np.zeros((2, 3), dtype=np.int64)
```

```
>>> a
array([[0, 0, 0],
       [0, 0, 0]])
```

```
# element-wise addition
```

```
>>> a + 1
array([[1, 1, 1],
       [1, 1, 1]])
```

```
# row-wise addition
```

```
>>> a + np.array([1, 2, 3])
array([[1, 2, 3],
       [1, 2, 3]])
```

```
# column-wise addition
```

```
>>> a + np.array([[4], [5]])
array([[4, 4, 4],
       [5, 5, 5]])
```

- Arrays don't have to have identical shapes to do calculations between them
- Smaller arrays are broadcasted to the larger shape
- Shapes need to be compatible as defined by the broadcasting rules

The broadcasting rule

Two arrays are compatible for broadcasting if for each *trailing dimension* (i.e., starting from the end) the axis lengths match or if either of the lengths is 1. Broadcasting is then applied over the missing or length 1 dimensions

More information and examples in the [documentation](#)

Detailed walk through examples

```
>>> a = np.zeros((2, 3), dtype=np.int64)
```

```
# row-wise addition
```

```
>>> b = np.array([1, 2, 3])
```

```
>>> b.shape
```

```
(3,)
```

```
>>> a + b
```

```
array([[1, 2, 3],  
       [1, 2, 3]])
```

```
# column-wise addition
```

```
>>> c = np.array([[4], [5]])
```

```
>>> c.shape
```

```
(2, 1)
```

```
>>> a + c
```

```
array([[4, 4, 4],  
       [5, 5, 5]])
```

- **a + b** : Axis 1 matches (3), axis 0 is broadcasted twice
- **a + c** : Axis 1 is has length 1, axis 0 matches (2), axis 1 is broadcasted three times
- More efficient than repeating arrays!

Advanced example: Outer product

```
>>> a = np.array([1, 2, 3])
>>> b = np.array([4, 5, 6])
>>> np.outer(a, b)
array([[ 4,  5,  6],
       [ 8, 10, 12],
       [12, 15, 18]])
```

```
>>> a.reshape(3, 1) * b.reshape(1, 3)
array([[ 4,  5,  6],
       [ 8, 10, 12],
       [12, 15, 18]])
```

```
>>> a.reshape(3, 1) * b
array([[ 4,  5,  6],
       [ 8, 10, 12],
       [12, 15, 18]])
```

- Here, broadcasting is used to calculate an outer product without using the `np.outer` function
- `a` is reshaped to a column vector
- `b` is reshaped to a row vector
- Broadcasting rules apply along both axes!
- `b` would be implicitly treated as a row vector, too. (axes are counted from the back)